研究報告

塑膠微粒對海洋生物的影響

可追溯的 LIVE evidence-bound research report

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塑膠微粒對海洋生物的影響的定義與範圍是什麼

The issue of microplastic pollution has garnered significant attention from governments worldwide and the academic community. [1]

OECD summarizes plastic leakage to aquatic environments and ocean pollution pathways, supporting policy framing for macroplastic and microplastic pollution reduction. [2]

海洋廢棄物問題日益嚴重，相當受關注，成為各國重要議題之一，為解決海洋廢棄物所產生問題，研擬相關計畫及資源，為能更有效減少海洋垃圾產生，透過調查臺灣周邊海域海漂底垃圾及微塑膠分布，掌握臺灣海洋廢棄物污染熱點。112年度共完成臺灣本島北部、中部、南部、東部及離島共5個區域 ... [3]

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塑膠微粒對海洋生物的影響的目前方法是什麼

Microplastics are now found throughout the world’s oceans, and although many organisms ingest microplastics, less is known about how plastics in seawater may affect key processes such as feeding rate, growth, and survival. [4]

GESAMP reviews sources, transport, fate and ecological effects of microplastics in the marine environment, including ingestion by marine organisms, food-web transfer and risk assessment needs. [5]

根據美國國家海洋暨大氣總署 (National Oceanic and Atmospheric Administration, NOAA)針對海洋廢棄物分解微小粒子 (體積尺寸小於5毫米之廢棄物)進行數量研究的相關分析資料顯示，有極大的占比為塑膠微粒 (Microplastics, MPs)，其中大多的塑膠微粒來自於大型塑膠碎裂而成，小部分來源則是各種塑膠原料。塑膠本身 ... [6]

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塑膠微粒對海洋生物的影響的工具與架構比較是什麼

Marine and coastal ecosystems are among the largest contributors to the Earth’s productivity. [7]

NOAA NCEI provides a global marine microplastics data product aggregating field observations into discoverable datasets for monitoring microplastic distribution in ocean and coastal environments. [8]

Evaluating the cumulative effects on coral growth from exposure to environmentally relevant microplastic concentrations and elevated temperatures Examining biological impacts to fish diets Related Research and Resources EPA's Trash Free Waters Program Microplastics Information from the National Oceanic and Atmospheric Administration [9]

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塑膠微粒對海洋生物的影響的評估與成效是什麼

UNEP assesses marine litter and plastic pollution, including microplastics, their sources, pathways, ecological impacts on marine organisms and ecosystems, and policy responses for monitoring and reduction. [10]

推動目的 聯合國訂定6 月8 日為世界海洋日(World Ocean Day), 辦理一系列海洋保護活動。2017 年世界海洋日主題為「我們的海洋、我們的未來(Our Oceans, Our Future) 」,保護行動的重點放在鼓勵解決塑膠污染和防止海洋垃圾。 [11]

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塑膠微粒對海洋生物的影響的限制與風險是什麼

Marine microplastic (MPs) pollution poses a significant threat to marine ecosystems and human health. [12]

Addressing MP pollution in the marine environment through a holistic framework—from identifying sources and characterizing their properties to evaluating biological accumulation and proposing stakeholder-based mitigation strategies—provides a comprehensive basis for effectively managing and preventing MP pollution in marine ecosystems . [13]

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塑膠微粒對海洋生物的影響的實務建議是什麼

隨著海洋塑膠污染問題日益嚴峻，微塑膠作為新型污染物已廣泛存在於熱帶近岸生態系統中，對生物多樣性與生態功能構成潛在威脅。本研究以海南沿海典型珊瑚礁生態系統為研究物件，系統調查微塑膠在海水、沉積物及不同營養級生物體內的豐度與類型，分析其空間分佈差異與生態遷移路徑。通過佈設多介質採樣點、實施穩定同位素與食物鏈追蹤實驗，結合生物攝食行為與顆粒理化特性，揭示了微塑膠在生態系統中的遷移規律及其潛在生態影響。研究發現，微塑膠在沿岸區域的濃度顯著高於外礁區，沉積物中以纖維狀和片狀顆粒為主，水體與底棲動物中則表現出明顯的環境回應性。中高營養級掠食性魚類體內的微塑膠累積量較高，間接證實其通過食物鏈逐級轉移的可能性。此外，生物的攝食策略、代謝能力與生態位元特徵在微塑膠遷移過程中發揮關鍵作用，不同物種對同類顆粒的回應存在顯著差異。儘管當前未形成典型意義上的生物放大效應，但微塑膠的長期生態風險不可忽視，尤其在珊瑚礁等生態敏感區更應引起高度警惕。基於研究結論，本文提出建立岸基—近海協同監測網路，加強塑膠源頭治理，構建資料驅動的微塑膠風險評估機制，為熱帶島嶼型生態系統的污染治理與生態保護提供實證依據與政策參考。 [14]

NOAA Marine Debris Program explains microplastics as small plastic debris in marine and aquatic environments, describing sources, persistence, monitoring concerns and potential effects on wildlife. [15]

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